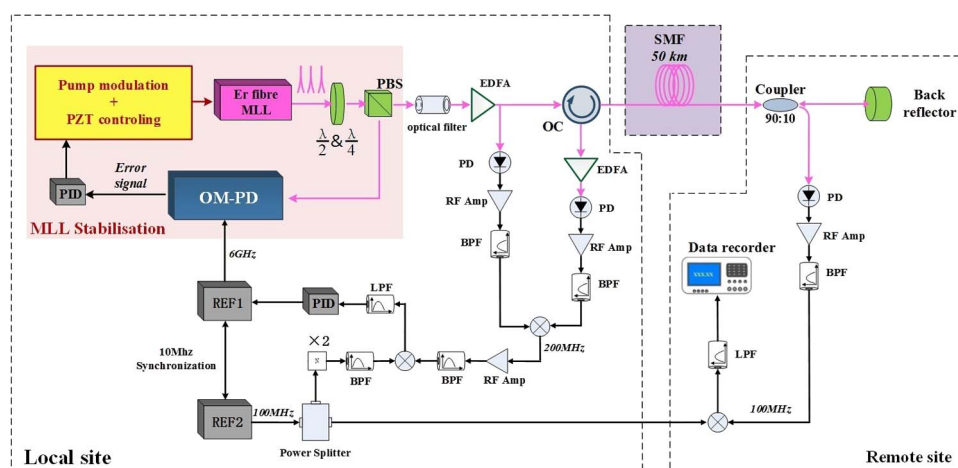


RF Transfer by Optical Comb Over 50-km Fiber Link With 4.7×10^{-18} Instability

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Abstract: A radio-frequency (RF) dissemination system using an optical comb over a 50-km fiber link has been demonstrated in this paper. An optical filter is utilized to suppress the dispersion influence of the optical comb in the transfer fiber link. The compensation scheme, based on an electrical compensation structure with an optical-microwave phase detector, cancels the phase fluctuation introduced by the temperature variation and the mechanical vibration in the transfer fiber link. It has been demonstrated that the relative frequency instability of a 100-MHz transfer signal is 1.9×10^{-14} at 1 s and 4.7×10^{-18} at 16 400 s. It indicates that our frequency dissemination system can be used to transfer the most precise RF signal in the world to meet the demand of multiply applications.

Index Terms: Radio frequency (RF), dissemination, optical comb, optical branching filter.

1. Introduction

Precise time and frequency dissemination is indispensable for human society in modern life. It has many significant applications, such as dissemination of the standard of world time [1], navigation system [2], remote sensing [3] and precise measurement. Currently, the most important significance of frequency dissemination is to synchronize the time and frequency of different sites by transferring the atomic clock. Numerous studies have been done to acquire a more accurate method of time and frequency dissemination. Over the past few decades, high precision transfer and synchronization of time and frequency over fiber link has aroused the interest of many researchers [4]–[28], because of its stability and precision, which can be as low as 10^{-21} order. Three main methods of time and frequency dissemination have been proposed, including optical frequency dissemination [4]–[10], microwave frequency dissemination [11]–[18], and optical frequency comb dissemination [19]–[22].

As the microwave signal is the most commonly used in daily life, various schemes of microwave frequency transfer over fiber link have been established. Microwave frequency transfer by amplitude modulating continuous-wave (CW) laser over fiber link is a well-established technique of microwave dissemination [11], [13], [15], and its instability reaches 4.5×10^{-19} /day [11] over 80 km fiber link. With this method, only a single microwave frequency point is transferred to the remote site. Nevertheless, microwave frequency can also be disseminated by optical comb [12], [17], [18]. In this way, microwave frequency signal is encoded in the

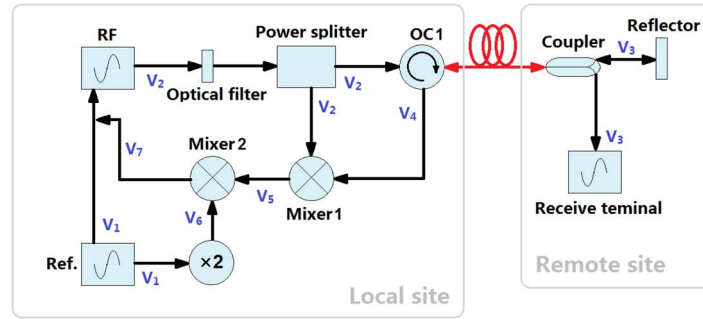


Fig. 1. Structure of RF transmission system (Ref: reference signal, OC: optical circulator.).

repetition rate of the optical comb, and the fundamental repetition frequency and its harmonic frequency can be acquired at the remote site. The instability reaches 6.5×10^{-19} level at 82500 s over 2.3 km fiber link [12]. However, with wide spectrum, optical comb would be severely dispersed over long fiber link. A technique to minimize dispersion is to utilize Dispersion-Compensating Fiber (DCF). DCF minimizes dispersion effectively [19], [21], [23], but it is unfit for general use given that the accurate matching is required in every fiber link. Since the communication fiber link is a single mode fiber and the transfer channel is divided by Wavelength Division Multiplexing (WDM) standard, it is very inconvenient to transfer optical comb with such a wide band. In our experiment, an optical filter with 0.5 nm band is introduced to narrow the band width of the optical comb. Thereby, the dispersion influence can be suppressed vastly in long distance fiber link. The repetition rate of optical comb would almost not suffer the dispersion. It is an effective and convenient method.

To obtain better instability, an electrical compensation scheme, which is of simplification and high quality [14], [24], is designed to compensate the phase fluctuation caused by the temperature variation and mechanical vibration. Besides, an optical-microwave phase detector (OM-PD) [12], [19], [25]–[27] is used to lock the repetition rate and the microwave signal with low additional noise. In our experiment, a 100 MHz frequency is transferred in a 50 km fiber link, and the instability is 1.9×10^{-14} at 1 s averaging time and 4.7×10^{-18} at 16400 s averaging time.

2. Theory and Principles

2.1. The Theory of Experiment Structure

The basic structure of the RF transmission system is showed in Fig. 1. The reference signal can be expressed as

$$V_1 = A_1 \cos(\omega t + \varphi_0). \quad (1)$$

V_2 is the transferred RF which has been encoded in the repetition rate of the optical comb. It is locked to the reference signal V_1 by an inner phase-lock loop 1 (PLL1) of the source. So the frequency of V_2 is consistent with the rate of reference signal ω . Hence, it inherits the phase of reference frequency φ_0 and introduces a constant phase delay φ_m . V_7 is the error signal of the out phase-lock loop 2 (PLL2), and is utilized to compensate the phase of V_2 by modulating the reference signal of V_2 . Therefore, the phase of V_2 contains a compensation phase φ_c . V_2 can be expressed as

$$V_2 = A_2 \cos(\omega t + \varphi_0 + \varphi_m + \varphi_c). \quad (2)$$

Carrying V_2 , optical comb passes through an Optical Branching Filter (OBF) to narrow the band of spectrum and decrease the influence of dispersion. The bandwidth of the optical filter is 0.5 nm and the center wavelength is 1550 nm. After a splitter, part of V_2 is transferred to the remote site over fiber link after an optical circulator. In this process, φ_d is the initial phase

delay in the transfer fiber link, and $\Delta\varphi$ is the variable part of phase delay, which is varying as time. It is induced by the variation of dispersion, which is further caused by temperature fluctuation and mechanical vibration. At the remote site, a signal V_3 can be given by

$$V_3 = A_3 \cos(\omega t + \varphi_0 + \varphi_m + \varphi_c + \varphi_d + \Delta\varphi). \quad (3)$$

Reflected by a Faraday Mirror, V_3 is transferred back to the local site along the same fiber link. After the OC1, the microwave signal V_4 can be given by

$$V_4 = A_4 \cos(\omega t + \varphi_0 + \varphi_m + \varphi_c + 2\varphi_d + 2\Delta\varphi). \quad (4)$$

An assumption is made that the phase noise is the same in the forward and backward process. V_5 is the up mixed signal of V_4 and V_2 , which can be expressed as

$$V_5 = A_5 \cos(2\omega t + 2\varphi_0 + 2\varphi_m + 2\varphi_c + 2\varphi_d + 2\Delta\varphi). \quad (5)$$

With V_1 to be the input of the frequency doubler, the output signal of the frequency doubler is given by

$$V_6 = A_6 \cos(2\omega t + 2\varphi_0). \quad (6)$$

V_5 and V_6 are both the inputs of Mixer 2. We can get V_7 the result of down-conversion of V_5 and V_6 , which could be expressed as

$$V_7 = A_7 \cos(2\varphi_m + 2\varphi_c + 2\varphi_d + 2\Delta\varphi). \quad (7)$$

V_7 is the error signal needed to control the compensation phase φ_c to suppress the $\Delta\varphi$: the varying part of phase delay in the fiber link, which means that $\varphi_c = \Delta\varphi$. At the remote end, the signal V_3 will be immune to the phase fluctuation.

2.2. The Analysis of Dispersion Influence

There is a severe dispersion when optical comb is transferred in fiber link between a long distance. The width of the pulses increased observably. In consideration of second-order dispersion, the pulses width of the remote location T_1 can be expressed as [28]

$$T_1 = \left[\left(T_0 + \frac{C\beta_2 z}{T_0} \right)^2 + \left(\frac{\beta_2 z}{T_0} \right)^2 \right]^{\frac{1}{2}}. \quad (8)$$

T_0 is the width of local pulses, z is the length of the transfer fiber link, β_2 ($\beta_2 < 0$) is second-order dispersion coefficient, and C is a constant. The original local optical comb is approximately Gauss pulse, and the C is 0. It is obvious that the pulses width of the remote location T_1 would get larger as the transfer distance z is increased. If a filter is utilized, a positive chirp is induced and $C > 0$. Thus, while z is increased, the increasing rate of T_1 will be reduced.

Our simulation method is based on the above analyzing [28]. The intensity of local pulse with 150 fs width is normalized to 1, β_2 is $-18 \text{ ps}^2/\text{km}$, and z is 10 km, 20 km, and 50 km. The result of remote pulse is showed in Fig. 2(a). It is demonstrated that the remote pulse turn out to be quite low and wide, and get worse caused by dispersion if the transfer distance is longer. Then, frequency comb with 1550 nm center wavelength, 100 MHz repetition rate and 150 fs pulse width is simulated to be transferred in a 50 km fiber link. In Fig. 2(b), the result shows that the normalized intensity decreases severely and the pulses overlap with each other totally. In this case, it is hard to get accurate signal. If the optical comb is filtered with 1550 nm center wavelength and 0.5 nm bandwidth and normalized to 1 before being transferred, the simulation result is shown in Fig. 2(c). Compared with no filter, the shape of frequency comb does not change that much. Therefore, it is a useful method to extract accurate repetition rate of the optical comb.

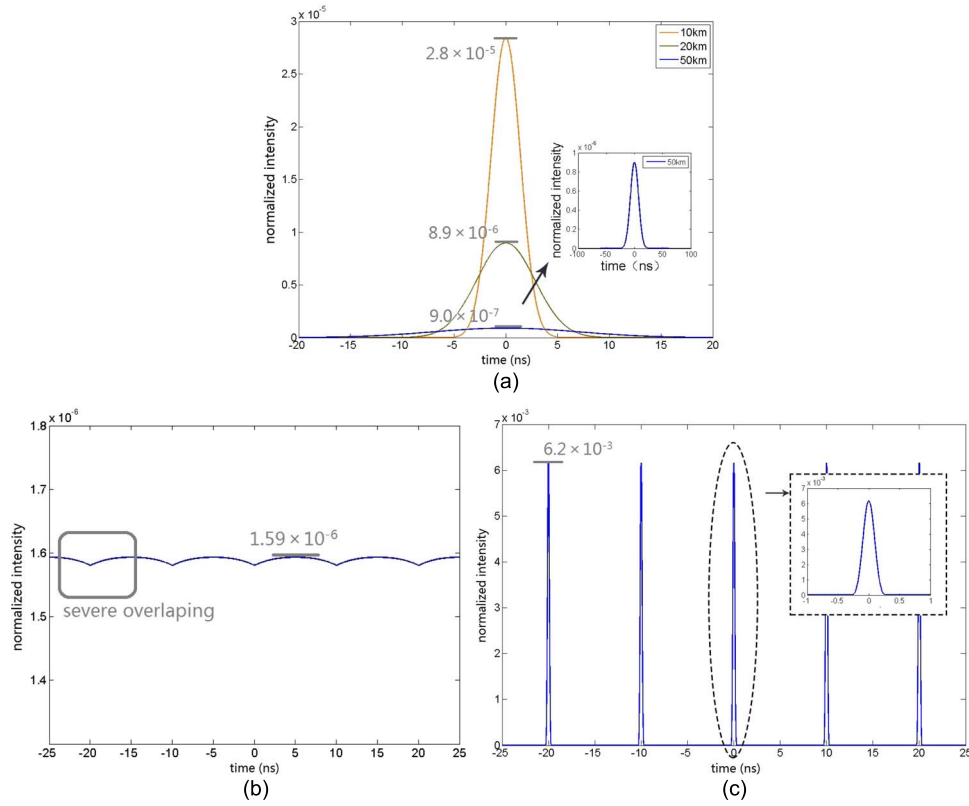


Fig. 2. Simulation result. Pulse width = 150 fs, normalized intensity = 1, repetition rate = 100 MHz, $\beta_2 = -18 \text{ ps}^2/\text{km}$. (a) Transfer result of a single pulse over 10 km, 20 km, and 50 km fiber. (b) Transfer result of optical comb over 50 km fiber. (c) Transfer result of optical comb with filter over 50 km fiber.

When $T_0 < 1 \text{ ps}$, third-order dispersion can also not be ignored. The pulse width of the remote location T_2 can be expressed as

$$\frac{\sigma_2}{\sigma_0} = \left[\left(1 + \frac{CB_2 z}{2\sigma_0^2} \right)^2 + \left(\frac{\beta_2 z}{2\sigma_0^2} \right)^2 + (1 + C^2)^2 \frac{1}{2} \left(\frac{\beta_3 z}{4\sigma_0^3} \right)^2 \right]^{\frac{1}{2}} \quad (9)$$

$$\sigma_2 = \frac{T_2}{\sqrt{2}} \quad (10)$$

$$\sigma_0 = \frac{T_0}{\sqrt{2}}. \quad (11)$$

β_3 is third-order dispersion coefficient. The pulse width of the remote location T_2 would also be broadened severely, while second-order dispersion and third-order dispersion are both considered. In our simulation, β_2 is $-18 \text{ ps}^2/\text{km}$, β_3 is $0.2 \text{ ps}^3/\text{km}$, pulse width is 150 fs, and normalized intensity is 1. Fig. 3(a) shows the results of the pulse transferred over a 50 km fiber and a 55 km fiber. Besides the T_2 gets larger, it also indicates that the top of the pulse drifts, if the transfer distance changes. In the transfer system, the length of optical path is influenced by temperature and vibration; therefore, it would cause pulse jitter. In Fig. 3(b), if we utilize a filter with 0.5 nm bandwidth before the pulse transferred in the simulation, pulse drifts get smaller. Therefore, the instability can be highly improved with a filter.

3. Experimental Setup

Based on the above analysis and illustration in Fig. 4, a transmission system is set up. At the local site, there are two reference frequencies which are showed as REF1 and REF2. REF1

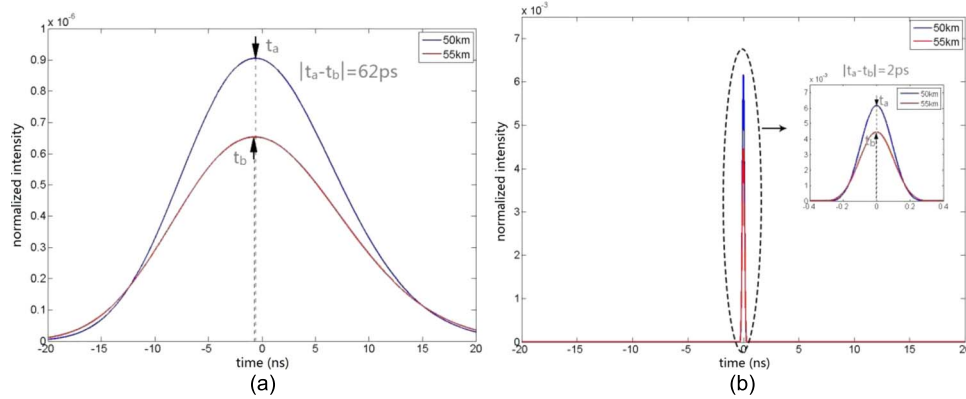


Fig. 3. Simulation result. Pulse width = 150 fs, normalized intensity = 1, repetition rate = 100 MHz, $\beta_2 = -18 \text{ ps}^2/\text{km}$, $\beta_3 = 0.2 \text{ ps}^3/\text{km}$. (a) Transfer result of a single pulse over 50 km and 55 km fiber. (b) Transfer result of a filtered single pulse over 50 km and 55 km fiber.

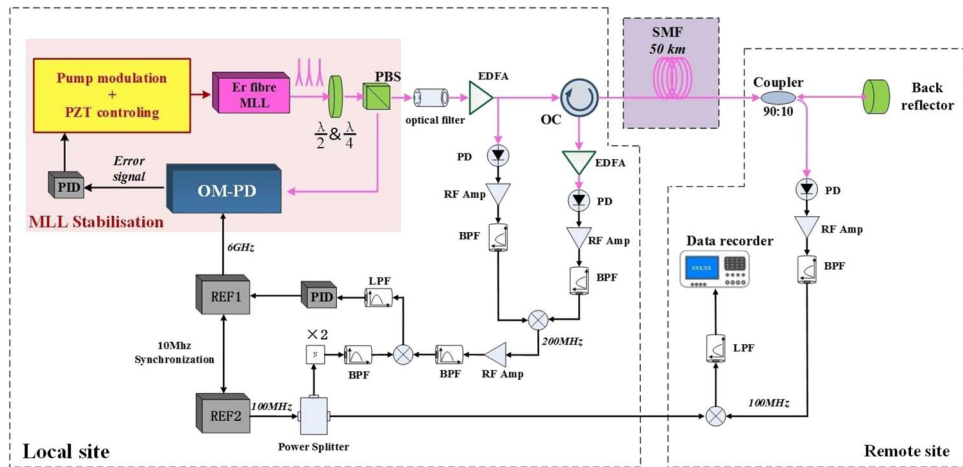


Fig. 4. Experimental setup of RF transmission system. (REF: reference signal, OC: optical circulator, PID: proportional-integral-derivative control, OM-PD: optical-microwave phase detector, MLL: mode-locked laser, SMF: single mode fiber, EDFA: erbium-doped fiber amplifier).

is synchronized to the 10 MHz inner frequency of REF2. REF1 is the reference signal of the repetition rate of the optical comb. It would be modulated by error signal. REF2 is split to two parts. One part is doubled by a multiplier and used as the LO of a mixer to extract the error signal of the fiber link. The other part is used to evaluate the out-of-loop instability of the transmission system.

An Er-doped fiber mode locked laser is used to generate a 1550 nm optical comb transferring in the fiber link. The repetition frequency of the optical comb is 100 MHz. The light is divided into two parts by a PBS. One part is used for frequency transfer, while the other is used to lock the repetition rate of the optical comb. In order to balance the two parts with appropriate intensity, a half-wave plate (HWP) and a quarter-wave plate (QWP) are induced before the PBS. To synchronize the repetition rate of the optical comb and the RF signal with low additional noise, an optical-microwave phase detector (OM-PD) is utilized [26], [27]. The output of the OM-PD is the error signal between the harmonics of the repetition rate and a 6 GHz RF signal generated by a VCO. To obtain more accurate locking result, a proportional-integral-derivative (PID) module is utilized, which is widely used in the field in electronic engineering. After a PID module, the error signal is used to adjust the pump and curve length via piezoelectric ceramic transducer (PZT) of the mode locked laser. In this way, the repetition rate is locked to the RF signal.

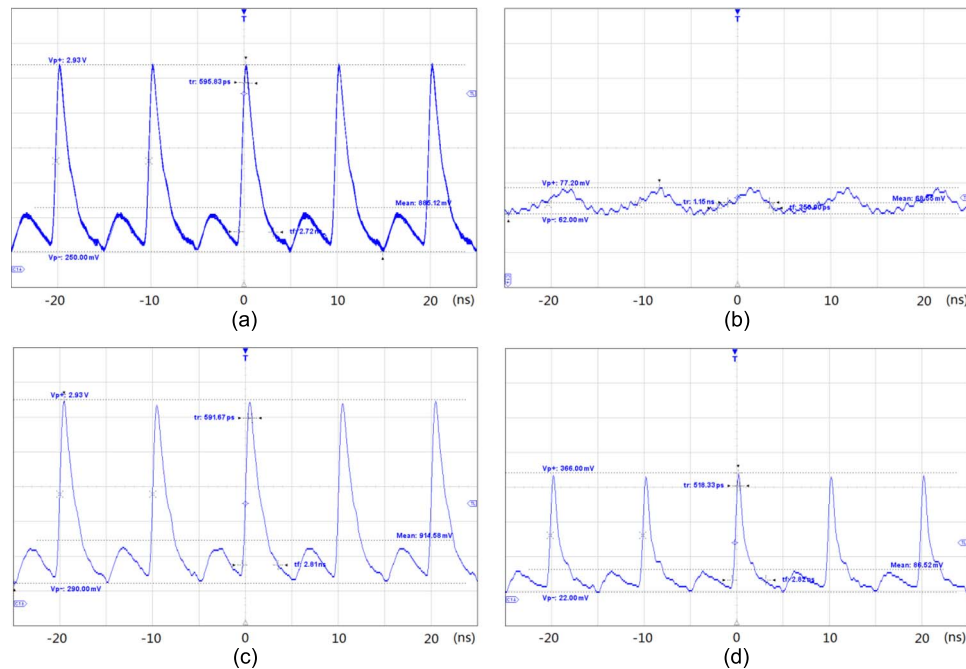


Fig. 5. (a) Local detected signal of optical comb. (b) Remote detected signal of optical comb. (c) Local detected signal of filtered optical comb. (d) Remote detected signal of filtered optical comb.

If a wide band optical comb were transferred, severe dispersion would be induced to deteriorate the transfer signal. With an Optical Branching Filter (OBF) to get a narrow band optical comb signal, the effect of dispersion is reduced obviously, which has been illustrated in Section 2. However, the most optical power of optical comb is lost due to the OBF. An erbium-doped fiber amplifier (EDFA) is used after the OBF. After amplification, the optical power is 14.5 dBm. Then it is split in two parts. One part is left at the local site as the LO of the up-mixer to extract the error signal of the fiber link. The other part is transferred through the fiber link. At the remote end, 10% of the light is left while the major is reflected by a Faraday Mirror to the local site. However the backward light is so weak that an EDFA is required to amplify the light to a proper level. After that, both the two arms of light are detected by photodetector respectively. Then the RF signals are demodulated. In order to get the error signal with minimum influence of the mixer, the input power of the mixer should be stringently controlled. Hence the amplifiers with proper gain are used. After the amplifiers, band-pass filters (BPF) are used to filter the irrelevant signals. Finally, 200 MHz frequency signal, the sum of the two arms is obtained. After a 200 MHz BPF, it is mixed with the doubled signal of the RF signal and the error signal is obtained. After a low-pass filter (LPF) with 1 KHz bandwidth, the error signal is used to compensate the phase fluctuation by adjusting the REF1. In order to realize a long term and robust control, another proportional-integral-derivative (PID) module is induced in this phase-lock loop 2 (PLL2).

The out of loop instability is analyzed by mixing the reference signal and the signal at the remote end. The measurement bandwidth is 1 KHz.

4. Results

To demonstrate the effect of optical filter on the dispersion in the 50 km fiber, the signals at local site and remote site are recorded. The optical comb is 8.3 dBm, with 100 MHz repetition rate and 150 fs pulse width. Fig. 5(a) is the local signal detected by photodetector, and the filter is not used. Fig. 5(b) is the remote signal detected by photodetector. It is indicated that the peak

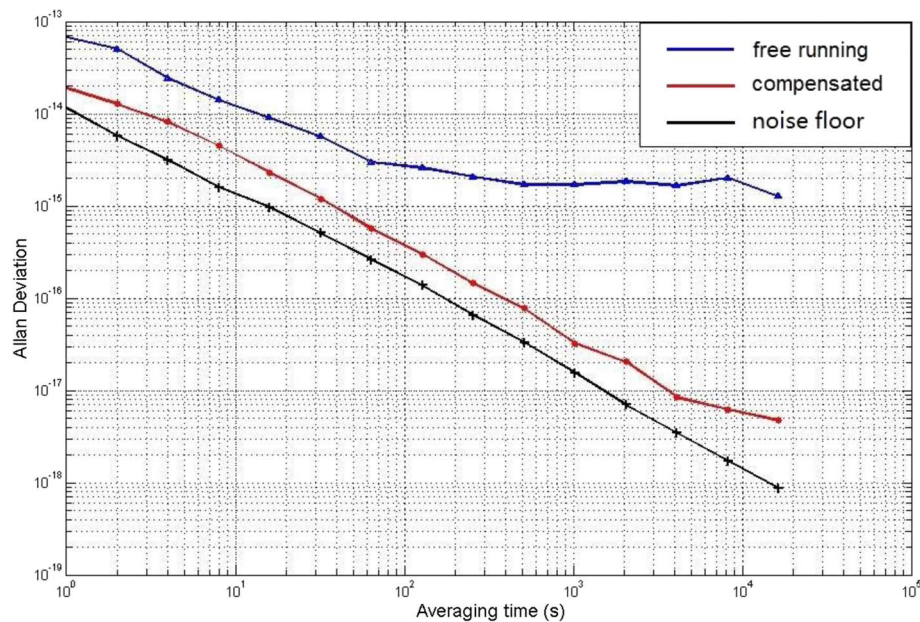


Fig. 6. Result of residual frequency instability. The blue curve is the instability of remote RF without compensation. The red curve is the instability of remote RF with compensation. The black curve is the noise floor of the system.

power decreases to at most 2.6%. The signal transferred over 50 km fiber cannot be recognized easily, and the optical comb becomes malformed. It is disadvantageous to extract accurate repetition rate of optical comb. When an optical filter with 1550 nm center wavelength and 0.5 nm bandwidth is utilized at the local site, the power of optical comb becomes -11.1 dBm. After an erbium-doped fiber amplifier (EDFA), the power is amplified to 8.3 dBm. Fig. 5(c) and (d) show the local signal and the remote signal detected by photodetector, respectively. It is showed that the peak power of the pulse decreases to 12.5%. The shape of the detected signal and the electrical pulse width nearly have no change. Limited by the upper response of our photodiode, the peak of electrical pulse is up to 3.5 V. The response of photodetector is also not linear when optical pulse has large power. This is the reason why the dispersion and peak power attenuation in simulation seems larger than experiment. With the comparison of Fig. 5(b) and (d), if an optical filter is utilized, it is easier to get precise repetition rate, and the dispersion is not that severe.

At remote site, we utilize the method mentioned in Section 3 to acquire the residual frequency instability. The compensation system could run over 10 hours. A digital multimeter is used to record the output voltage of the mixer which containing the information of the out of loop instability. Then the data is processed and the instability performance is expressed by the Overlapping Allan Deviation, which is showed in Fig. 6. The residual frequency instability with the compensation scheme could reach 1.9×10^{-14} at 1 s and 4.7×10^{-18} at 16400 s, while the residual frequency instability is 6.7×10^{-14} at 1 s and 7.6×10^{-16} at 16400 s for the free running system. The black curve shows the noise floor of the system. It mainly reflects the limit of the system.

It can be easily recognized that with the compensation scheme the residual frequency instability would decrease obviously compared with the free running system. The instability decreases half order at 1 s and over 3 orders at 16400 s. It indicates that this compensation scheme improves long term stability primarily to eliminate the influence of the phase fluctuation induced by the fiber link.

However it is not consistent with the noise floor curve due to the compensation structure of its own. A fundamental assumption of the phase compensation structure is that the forward and

backward phase noise induced by the environment are the same. The forward phase fluctuation in the fiber link is equal to the backward, but the error signal extraction structure and phase pre-compensation structure at the local site may induce weak asymmetric phase noise. This leads to the inconsistency of red curve and black curve in Fig. 6.

It is another method to utilize Dispersion-Compensating Fiber (DCF) to compensate dispersion. Some research has been done in other work. The transfer instability of 6×10^{-17} at 1000 s is reported between 43 km distance [23]. Our system has improved the instability to about 3×10^{-17} at 1000 s over 50 km fiber link. The instability could reach 6×10^{-18} at 1000 s [21] and 9×10^{-18} at 1000 s [19] by using DCF, but the transmission distance is only 7.7 km [21] and 10 km [19]. DCF compensation scheme is mainly researched in short distance transfer, because DCF is quite expensive. In the Wavelength Division Multiplexing (WDM) standard, the channel is 1 nm or narrower. DCF will not narrow the spectrum of optical comb to this order. Meanwhile, it is another difficult work that the length of DCF must match the single mode fiber accurately. Our compensation system is convenient for optical communication comparatively. Optical Branching Filter (OBF) is a common component and not expensive. The spectrum width of filtered optical comb is just suit for optical communication. This method can reach the same order instability of DCF compensation scheme.

5. Summary

We propose a RF dissemination system using optical comb over 50 km fiber link. To suppress the dispersion influence on optical comb in the single mode fiber, an optical filter is utilized to narrow the bandwidth of optical comb. An improved error signal extraction method is used in the phase compensation structure. In addition, an optical-microwave phase detector (OM-PD) is utilized to improve the locking property between the repetition rate of Er-doped fiber mode locked laser and reference RF. It has been demonstrated that the relative transfer stability can reach 1.9×10^{-14} at 1 s and 4.7×10^{-18} at 16 400 s. With this compensation scheme, fluctuation of transfer fiber link could be compensated.

It demonstrated an improved performance compared with the previous work [29]. More importantly, it provides an available approach to disseminate the most precise microwave standard to the remote end with multiple frequencies.

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